NNET can achieve a slightly higher BA. Nevertheless, the computational cost for training an NNET is significantly higher than that of QDA, which also provides good results. For example, the contour lines in Fig. 3.1 (b) are based on QDA, which achieves the highest BA for Olea europaea.

3.2 MCMC performance

In addition to the isotope data from the Mediterranean Sea (Medstack), from globally distributed ocean cores (LR04) and from Greenland (NGRIP) described above, proxy data from various lake and mire sediments and from caves are also used in this study. Specifically, these are the pollen records and the speleothem isotope data from the caves shown in Fig. 3.3. If a Bayesian age-depth model can be calculated for a sediment profile from the European Pollen Database, both MCMC methods are applied to it. These are the Metropolis-within-Gibbs (MG) and random walk Metropolis-Hastings

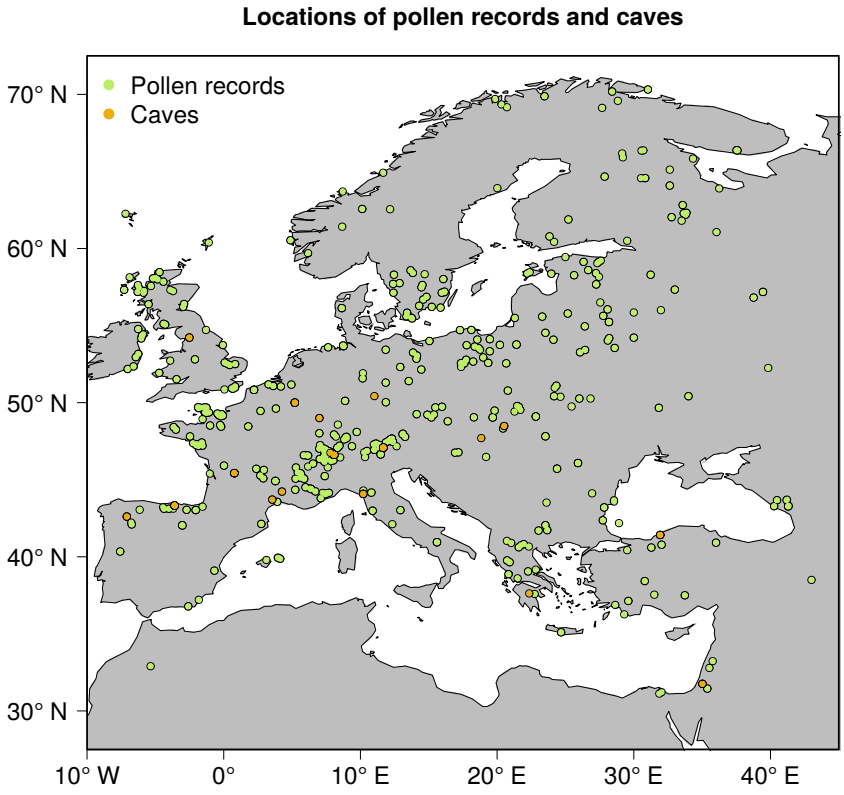


Figure 3.3: This map shows the locations marked with green dots where MCMC simulations are performed. These describe a subset of the European Pollen Database and include fossil and modern pollen records from natural archives such as lake and mire sediments. The brown dots show the caves where we take isotopic information from speleothems. These are a subset of SISAL.

algorithms. If caves are within 150 km of a given location with pollen records, our reconstruction tool compares them according to the lowest mean age resolution, the highest common age range of sediment and speleothem, and the shortest distance between them. Finally, an entity is chosen via a corresponding Pareto optimum. This describes the best possible state of all the characteristics mentioned, whereby it is not feasible to improve one of the characteristics without simultaneously worsening another (Langenbrunner and Neelin, 2017). In our case, the brown dots in Fig. 3.3 show the caves used in this work. The respective data come from the speleothem database SISAL.

Our study includes a total of more than 600 MCMC simulations. Below we summarize some important features of these final results by looking at the most common MCMC indicators. To this end, all simulations are first examined and then a reconstruction of Lake Prespa is considered in more detail.

3.2.1 All simulations

Fig. 3.4 shows some insights into all MCMC simulations. First, it is noticeable that the average computation time in (b) is higher than in (a). This is caused by an increase from 12,500 (MG) to 50,000 (rwMH) iterations and is necessary because of the higher integrated autocorrelation time (IAT) shown in (g) and (h). IAT is an estimate of the average number of iterations for drawing an independent sample in a Markov chain (Christen and Fox, 2010). In addition, two further MCMC diagnoses are given, namely the Gelman-Rubin convergence indicator (Brooks and Gelman, 1998) in panels (c) and (d) and the acceptance rate in panels (e) and (f). The first is calculated by running and comparing 10 simulations (chains) for each site and algorithm. More precisely, it measures the ratio of inter-chain to intra-chain variance. An example of a chain of 4 parameters is shown in Fig. 3.10 for Lake Prespa, which will be discussed in more detail later. Since we consider more than one parameter in the MCMCs, the multivariate extension of this type of analysis of variance (ANOVA) is calculated as described in Brooks and Gelman, 1998. The closer this ratio is to 1, the more likely it is that convergence has been achieved. Gelman et al., 2013 recommend a ratio of less than 1.1, which is guaranteed in all simulations. Moreover, both medians are quite close to one (two decimal places are shown), and only a few are above 1.04. The acceptance rate in (e) and (f) is the second MCMC diagnosis and shows higher values on average for the MG. Note that the Gibbs part of this method has an acceptance rate of 1, but the MG part can be checked in the same way as with rwMH. In panel (e) not only the median is higher, but also the variance. The reasons for these differences are explained in the following.

Since Gibbs sampling in our case uses the categorical distribution to determine the taxa weights (see Eq. 2.26), only a finite number of different taxa combinations is possible, unlike rwMH. More precisely, the categorical distribution requires, in addition to the weights, a parameter indicating the number of taxa considered. We choose half the number of taxa because $\binom{K}{t}$ is maximal when $t = K/2$. In this case, K is the number of taxa considered and t is the parameter inserted into the

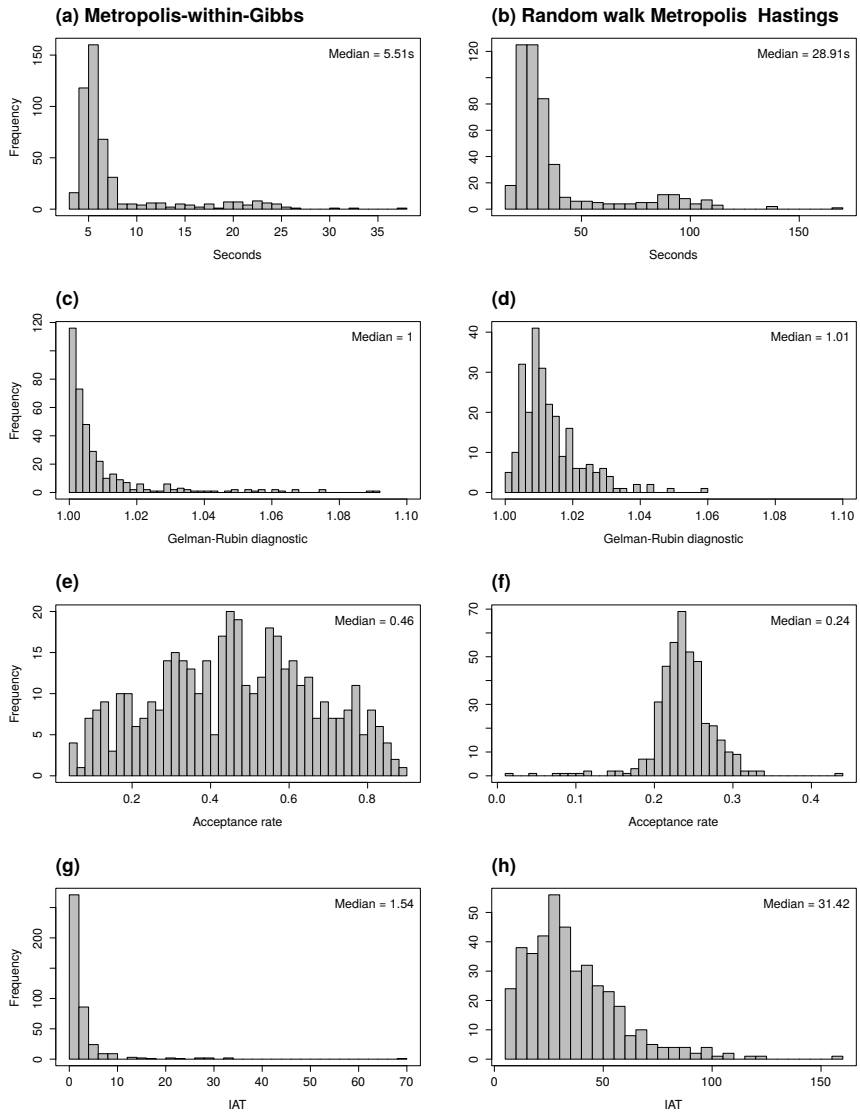


Figure 3.4: The left column shows MCMC results based on Metropolis-within-Gibbs sampling and the right column the results based on random walk Metropolis-Hastings. Panels (a) and (b) depict the computation time of each simulation and (c) and (d) a convergence indicator based on the Gelman-Rubin convergence diagnosis. In addition, panels (e) and (f) reveal the acceptance rate and (g) and (h) the integrated autocorrelation time of each MCMC simulation.

categorical distribution. On the one hand, this allows the algorithm to make larger "jumps" in the parameter space during the inference process, as omitting or adding different taxa can lead to larger changes in the information. The autocorrelation thus decreases faster than with rwMH. On the other hand, since not all taxa information are considered at the same time (as with rwMH), the step size can be relatively small. These extremes could lead to comparatively high and low acceptance rates (Lynch, 2007), which are shown in Fig. 3.4 (e). Note that each accepted reconstruction contains the

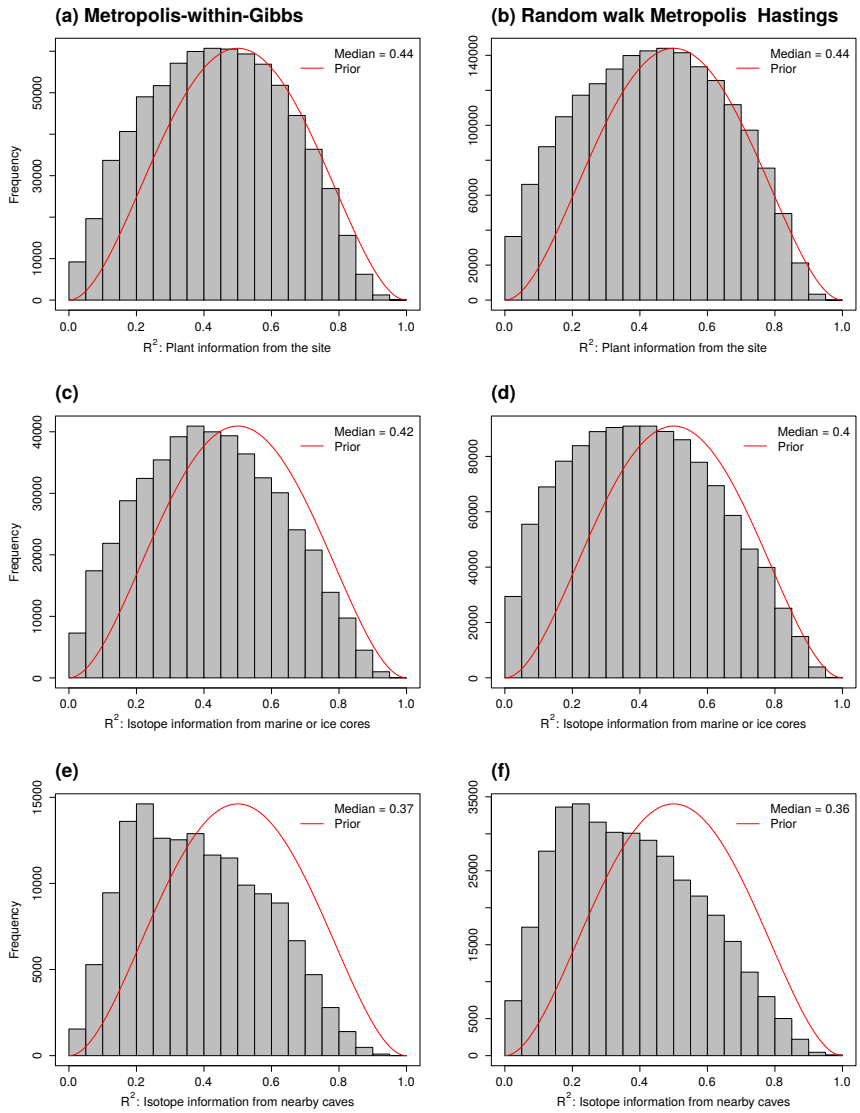


Figure 3.5: The left column shows MCMC results based on Metropolis-within-Gibbs sampling and the right column the results based on random walk Metropolis-Hastings. In summary, all explained variances of the final sample iterations are shown. Panels (a) and (b) depict R^2 with respect to plant information, (c) and (d) regarding isotopic information from marine or ice cores, and (e) and (f) concerning isotopic information from speleothems.

uncertainty information of the transfer functions. Therefore, even a single sample corresponding to our proposal distributions may be sufficient. However, with appropriate MCMC settings, we can still extract additional information from the proxies.

Fig. 3.5 and Fig. 3.6 give us an overview of all samples of all parameters. There are 5 parameters in total, namely the explained variances in terms of plant information, in relation to isotopic information from marine or ice cores, in reference to isotopic information from nearby caves, the annual temperature

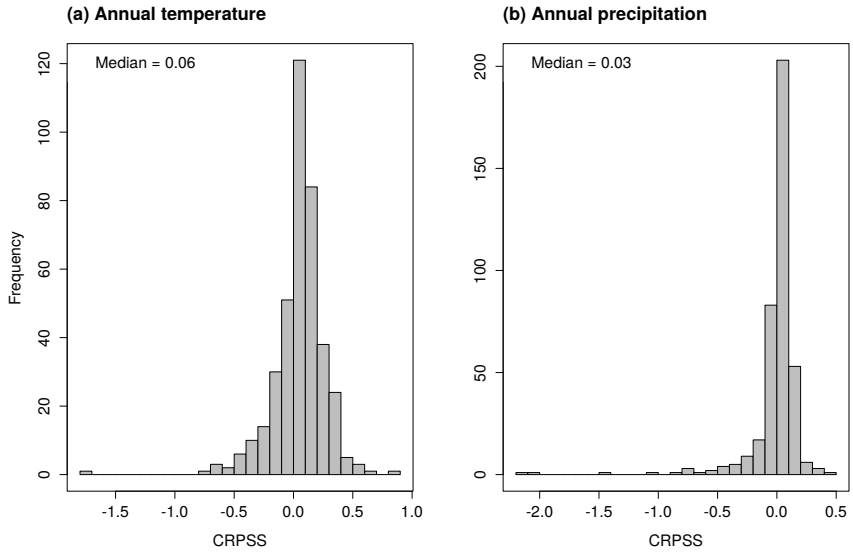


Figure 3.6: Continuous Ranked Probability Skill Score based on reconstructed time periods for ages younger than 1950 AD for annual temperature and precipitation compared to their respective current measurements. The "forecasts" are the results of the Metropolis-within-Gibbs method and the "reference forecasts" are selected from the random walk Metropolis-Hastings samples.

and the annual precipitation of the most recent time layers. Fig. 3.5 shows the explained variances for both methods and all MCMC simulations. The red lines represent the symmetric proposal prior distribution of R^2. Their means are around 0.5 and their variances are close to 0.2. These moments are used to find a compromise between the different locations and not to allow relatively high or low values. For easier comparison, the maximum values of the prior distributions shown correspond to the maximum values of the respective histograms containing the posterior samples.

In summary, the histograms for both methods look the same with respect to each parameter, which is also reflected in the medians. From top to bottom, the skewness of the posterior samples becomes more and more positive, indicating that the respective medians decrease. The explained variances in terms of the plant information are closest to the proposal distribution. This could be due to the fact that the taxa used for the reconstructions and the corresponding reference curves are based on the same sediment information (Giesecke et al., 2017). With regard to the reference curves from marine or ice cores, the values are somewhat lower. We see that the available taxa information for each core have sufficient variability so that these reference lines can be approximated in the manner presented. The lowest mean explained variance is found in panels (e) and (f), where pattern correlation of wavelet power spectra plays a role. Nevertheless, the medians indicate that the cores contain enough information to mimic the spectral patterns of nearby caves.

In Fig. 3.6 we see a comparison of the two methods in terms of annual precipitation and annual temperature. This is done by calculating the Continuous Ranked Probability Skill Score (CRPSS).

Specifically, the CDFs of the two climate variables from periods younger than 1950 AD are compared with the corresponding CRU data. Therefore, the Continuous Ranked Probability Score (CRPS) is calculated first, as described in Hersbach, 2000. To compare the two methods, we define the CRPS from MG as "forecast" and the CRPS from rwMH as "reference forecast". The CRPSS is then calculated via

$$\mathrm{CRPSS} = 1 - \frac{\mathrm{CPRS}_{\mathrm{forecast}}}{\mathrm{CPRS}_{\mathrm{reference\ forecast}}}. \tag{3.1}$$

This means that values close to 0 indicate that both methods have equal skill. We see that the medians of the two climate variables show that MG and rwMH have relatively similar posterior results. Except for some outliers, the histograms show distributions centered around 0.

All in all, both methods lead to similar results in parameter space. There are some dissimilarities in typical MCMC indicators due to differences in the inference of taxa weights. The MG has the advantage of converging much faster, but the mixing rate is sometimes a bit lower. Nevertheless, both methods lead to similar final quantitative reconstructions. An example of this is described in the next section.

3.2.2 Lake Prespa

We will now use the example of Lake Prespa to show how we determine a final reconstruction. Lake Prespa is located on the border between Greece, Albania and Northern Macedonia at an altitude of 849 m above mean sea level (amsl) (Aufgebauer et al., 2012). It is one of the oldest lakes in Europe and one of the three largest in the Balkan Peninsula. The lake has no surface runoff and drains into Lake Ohrid via karst aquifers. A quantitative paleoclimate reconstruction of Lake Ohrid is presented in Sect. 3.3.6. In Fig. 3.7 a climate diagram of the CRU grid point closest to Lake Prespa shows a maritime climate Cfb. Although drier climatic conditions prevail during the warm summer months, the Mediterranean influence is relatively low due to the orography and there is no year-round dry season. The mean annual temperature in this region is about 8.8 °C, and the average annual precipitation is around 705 mm. In this work, we use all CRU grid points of the Cfb climate classification to calculate the climate PDFs. In this way, the catchment area of Lake Prespa should be better represented than if only the nearest grid point is considered. This results in higher variance for both climate variables and provides our MCMC algorithms with higher flexibility in anchoring recent time slices within these distributions. Due to the orography, the climate is highly variable influencing the vegetation surrounding the lake, which act like an vegetation refuge during Quaternary climate cycles (Panagiotopoulos et al., 2020). Panagiotopoulos et al., 2013 describe the current vegetation as a mixture of Central European, Mediterranean, and Balkan endemic plants.

In autumn 2009, core Co1215 was drilled in the central northern part of the lake. The vegetation of the uppermost 320 cm, i.e., the last ca. 18,000 years, are examined in Panagiotopoulos et al., 2013. Later, additional samples were added to this palynological dataset to increase temporal resolution

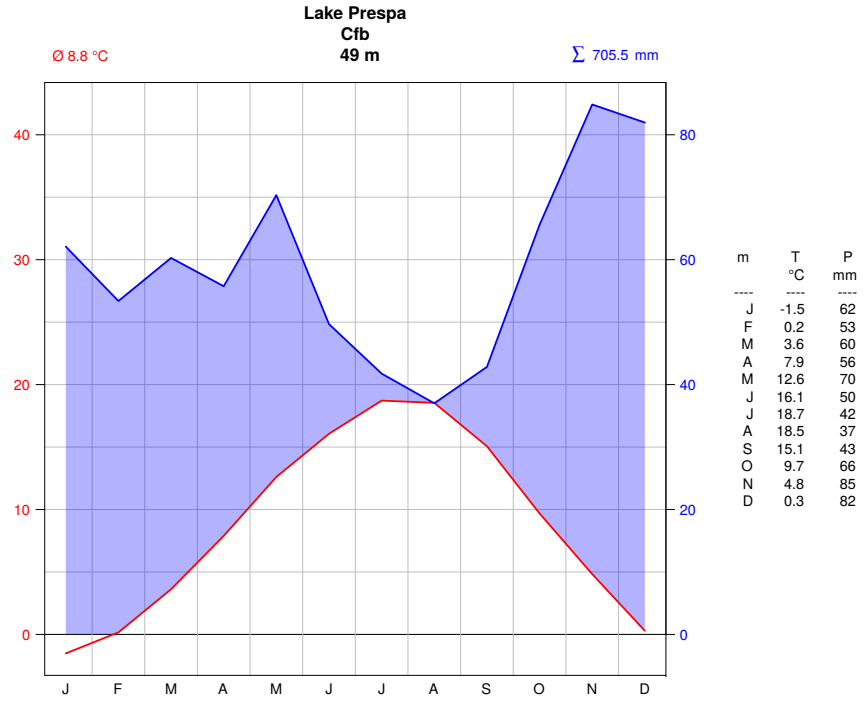


Figure 3.7: Climate diagram of the Lake Prespa region. The data are based on CRU 4.01 (Harris and Jones, 2017) and represent the grid point which is closest to the Lake Prespa over the period from 1961 – 1990 AD. The red colors and numbers represent the temperatures in °C and the blue ones the precipitation in mm. The letters on the x-axis indicate the months, which are also shown on the right side of the diagram, where the climate data are listed in a table.

(Konstantinos Panagiotopoulos, personal communucation). In Fig. 3.8 we see the two reference curves used for the climate reconstruction below. Panel (a) shows isotopic information from multiple cores in the Mediterranean Sea, referred to as Medstack (Colleoni et al., 2012). In (b) we see the percentage of arboreal pollen in relation to the percentage of non-arboreal pollen, based on the palynological data of the drill core. We correlate AP/NAP and Medstack respectively with the reconstructed annual temperature. This is based, on the one hand, on the discussion in Wagner et al., 2019, where Medstack is associated with warm/cold SST. On the other hand, Panagiotopoulos et al., 2013 link AP/NAP with warmer/cooler temperatures, respectively. As input to our Bayesian age-depth model, we use the tephra layers and radiocarbon ages described in Aufgebauer et al., 2012. The corresponding result is shown in Appendix A.

Let us first consider the acceptance rate of the two methods shown in Fig. 3.9 (a) and (b), which are generated during both MCMC simulations of Lake Prespa. The x-axes show the different number of iterations. We choose a burn-in size of 2,500 for MG and 25,000 samples for rwMH. Both curves show that the MCMC simulations seem to converge after these iterations. To account for autocorrelation (AC), every 5th iteration of the remaining samples is considered. Panels (c) and (d) contain the ACs of the final sample sizes of 2,000 and 5,000. This type of thinning is a good choice for MG because

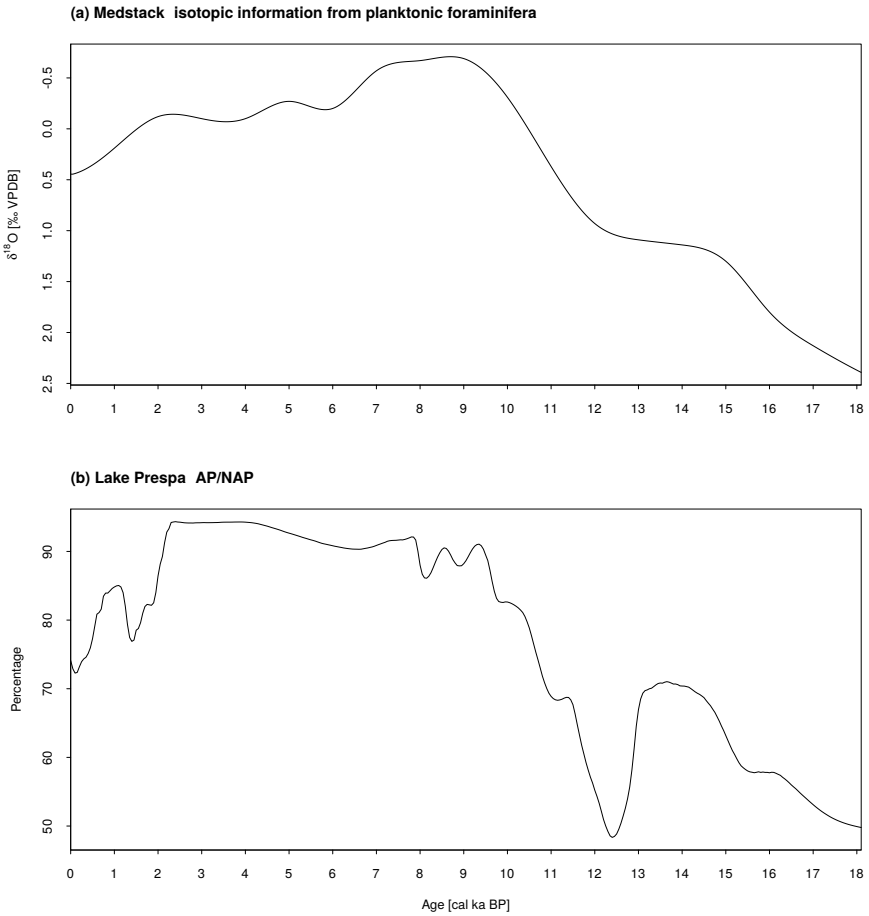


Figure 3.8: Reference curves based on (a) isotopic information from planktonic foraminifera from stacked marine cores of the Mediterranean Sea (Colleoni et al., 2012), and (b) on total terrestrial pollen sums: arboreal and non-arboreal pollen taxa from Lake Prespa (Panagiotopoulos et al., 2013).

the corresponding AC in (c) drops immediately after the first lag. In contrast, the rwMH method requires a larger thinning step. As already explained, the IAT in Fig. 3.4 (g) and (h) gives us an idea of such step sizes. Therefore, this indicator is chosen not only for further trimming of chains, but also to obtain independent posterior samples. First, however, we will look at examples based on thinning without IAT.

In Fig. 3.10 the thinned chains of the MG method are shown for 4 parameters. As SISAL does not offer a cave near Lake Prespa, this parameter is not included. In addition, panel (d) is based on isotope data from the Mediterranean Sea (Medstack). At this point it should be mentioned again that the flexible algorithm gives the users the possibility to include reference curves of their choice. On the right side we see the corresponding prior distribution and the posterior densities defined by each chain. For comparison, the posterior distribution of rwMH is also shown in the same diagrams, with all PDFs normalized to the same range. As far as the annual temperature in panel (a) is concerned,

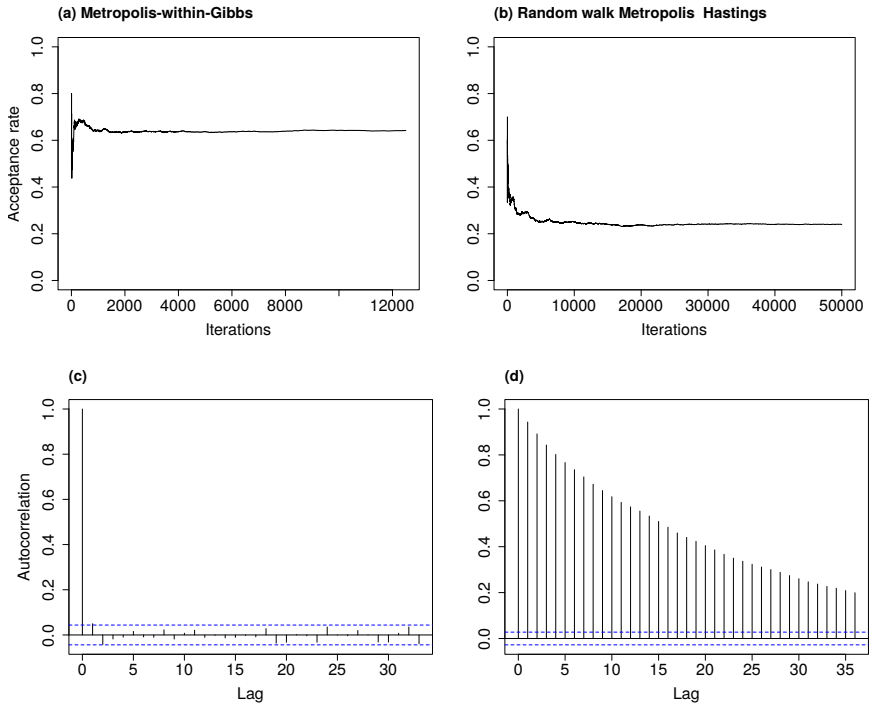


Figure 3.9: The left column shows the autocorrelation and acceptance rate of the Metropolis-within-Gibbs method for Prespa Lake. The right column depicts the same, but for the random walk Metropolis-Hastings method.

both posterior distributions look relatively similar. They have a higher mean and lower variance than the prior distribution. In (b) the output converges close to the mode of the prior gamma distributed annual precipitation. Here we see a bimodality in the MG method, indicating that the algorithm "jumps" between two convergence zones. However, the rwMH converges within the mode with higher precipitation. The explained variances with respect to the arboreal percentages from the core shown in (c) exhibit similar convergence behaviour for both methods, resulting in slightly higher posterior values. In comparison, the rwMH in (d) has a higher mean value than the MG. Both reference curves show a relatively high correlation in the case of Lake Prespa. Nevertheless, the additional computational cost is quite small and the final result contains, although not very much, more information than if only one reference curve is considered.

Even though there are slight differences in the results just explained, the overall reconstruction of the two methods is relatively similar. This is shown in Fig. 3.11. Depicted are the PDFs of the mean values derived by the MCMC parameters described above for both methods. The only obvious difference can be seen for ages older than 12 cal ka BP in panel (c), where there are two branches of annual precipitation. This is also reflected in the bimodal posterior distribution of MG in Fig. 3.10 (b). The comparison between MG and rwMH shows that the mode with higher annual precipitation for

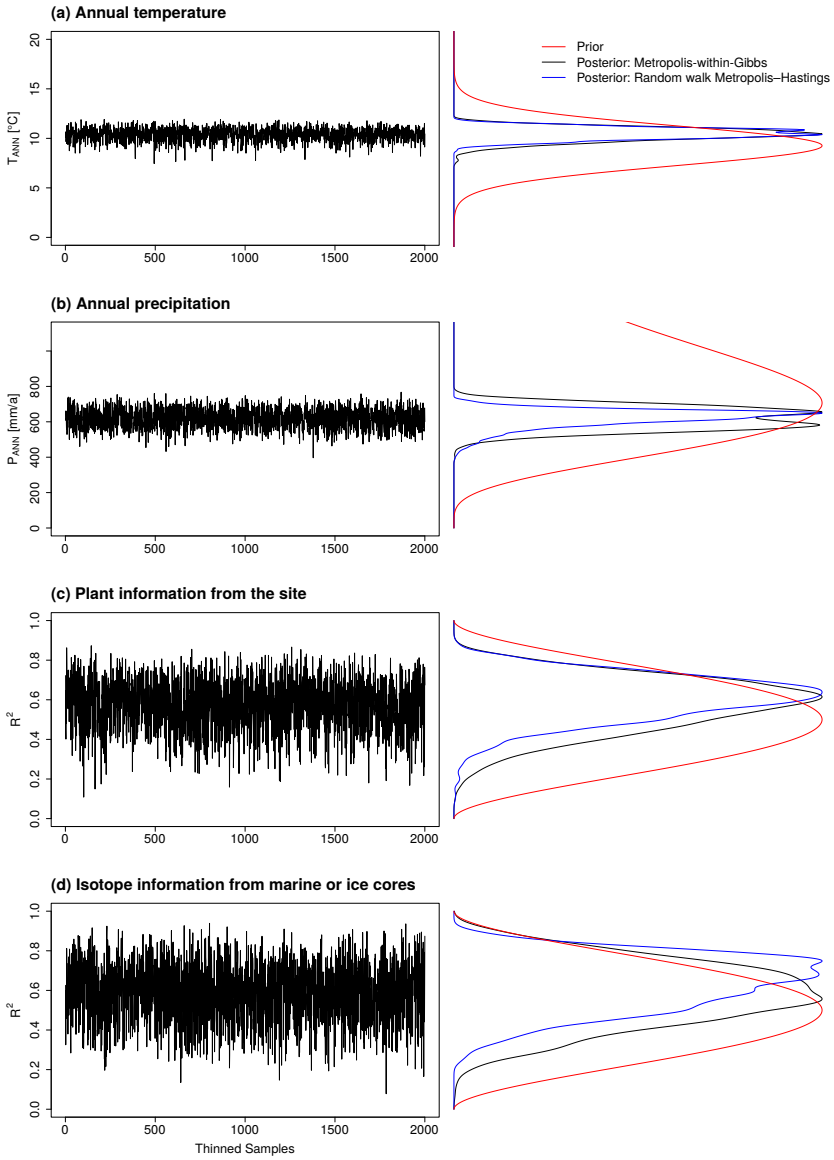


Figure 3.10: Trace plot and corresponding PDFs of the MCMC simulation of Lake Prespa. The left column shows the thinned samples from the Metropolis-within-Gibbs method for both climate variables and the explained variances with respect to the plant and isotope information. The right column depicts the PDFs based on these samples from both MCMC methods, distinguished by black and blue lines. The red lines show the respective prior distribution for each parameter.

the most recent time slice leads to the branch with lower precipitation in the late Last Glacial (not shown). Ultimately, however, this difference does not affect the common trade-off between the other parameters, as panels (a) and (b) do not show large discrepancies.

So far, we have only examined the mean values of the reconstructions. However, for a final climate reconstruction, we want to include the uncertainty information that results from the probabilistic

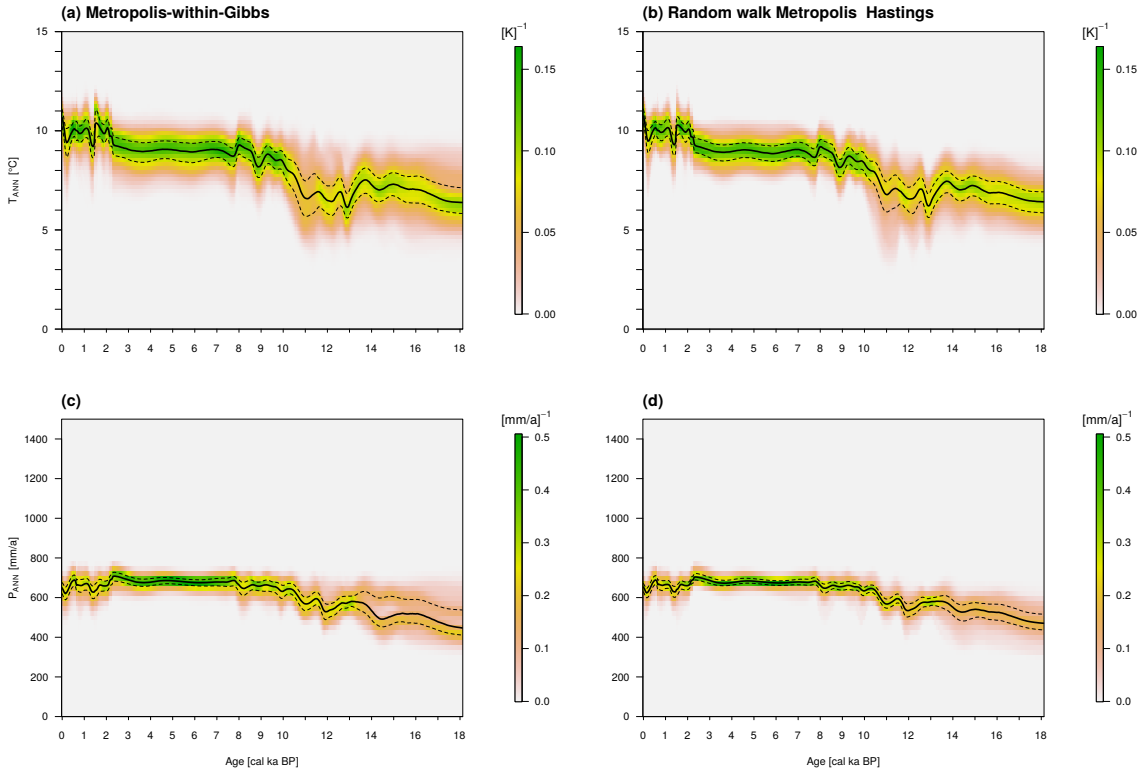


Figure 3.11: Posterior PDFs of all medians of the two climate variables (first and second row) and for both methods (first and second column).

transfer functions. There are many ways to do this, and we will briefly introduce some of them below and finally present the most appropriate one for our concern.

As a first step, the chains in Fig. 3.10 are first pruned by selecting those samples that lie within the interquartile ranges of the individual prior distributions. In a second step, trimming is continued until the IAT is below 1.3. Finally, a Pareto optimum is used to select the samples closest to the corresponding prior means. Here the user can define how many samples are to be selected at the end. Initial tests show that 10 individual reconstructions contain enough variability to describe large parts of the MCMC simulation. For this reason, 10 samples are always considered for each quantitative climate reconstruction in the following discussions.

The Figs. 3.12, 3.13, and 3.14 show different ways in which these 10 final samples can be combined. Note that each sampled reconstruction contains the uncertainty information from the transfer functions. So we have to deal with the combination of PDFs. Each of the following calculations is performed separately for the two climate components of the vector $\boldsymbol{C}$ and is also true for the corresponding random vectors $\mathcal{C}_i$. For simplicity, the two expressions are not presented decomposed into their respective components. Fig. 3.12 shows the final mixture PDF $\mathbb{P}_{\text{mixt}}(\boldsymbol{C})$. These weighted averages are

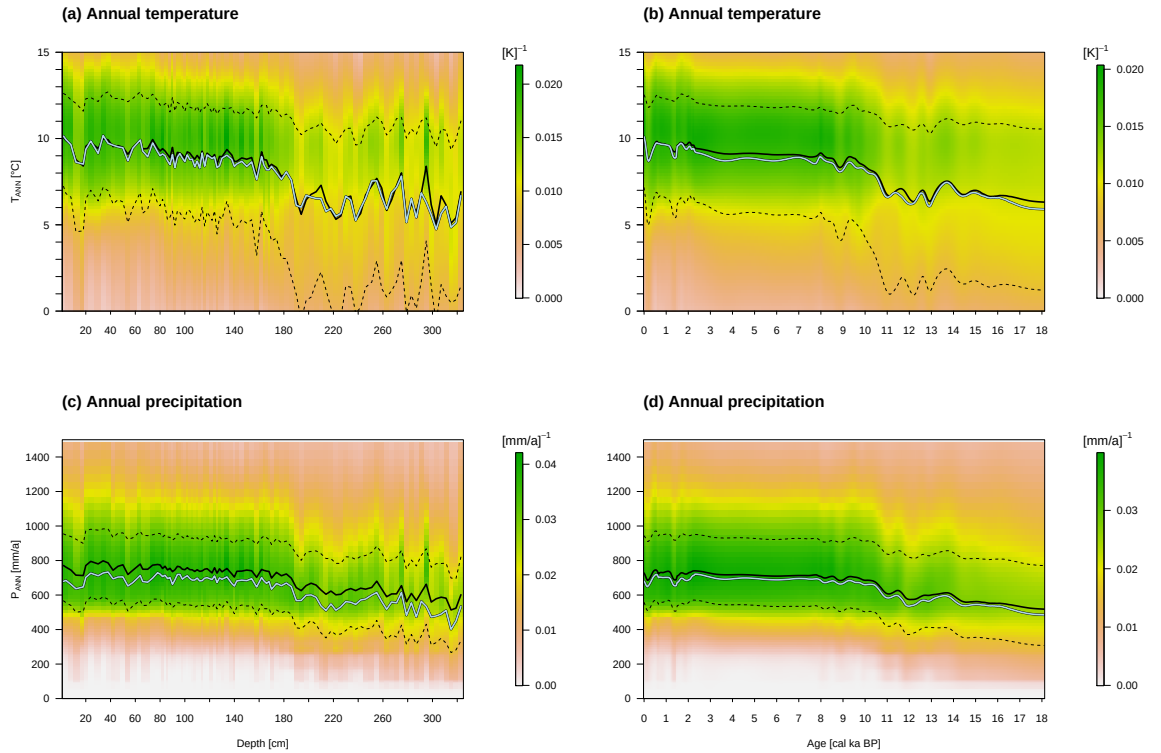


Figure 3.12: Posterior PDFs of the two climate variables (first and second row) and depth (first column) and age (second column). The basis of this result is the weighted average of all posterior PDFs. The black lines are the medians of the colored probability density values calculated using the Metropolis-within-Gibbs results. The blue lines are based on the random walk Metropolis-Hastings method.

calculated as follows:

$$\mathbb{P}_{\text{mixt}}(\boldsymbol{C}) = \frac{1}{N}\sum_{i=1}^{N} w_i \cdot \mathbb{P}_{\mathfrak{e}_i}(\boldsymbol{C}); \quad \text{with: } w_i > 0 \text{ and } \sum_{i=1}^{N} w_i = 1. \tag{3.2}$$

$N = 10$ is the number of independent climate reconstructions and w the corresponding weights. Here, each user can decide for themselves which weights they want to pass on to the algorithm. These could be defined, for example, by the lowest mean variances of the reconstructions or the highest mean probability values with respect to the prior PDFs. In this study, we use the same weights for each PDF. To demonstrate the age-depth transformation in relation to these PDFs, we plot the depths in the left column. The advantage of Eq. 3.2 is the low computational cost, that the input PDFs do not have to be independent and that the mean of $\mathbb{P}_{\text{mixt}}(\boldsymbol{C})$ (black and blue solid lines) is the average of the means of the input distributions $\mathbb{P}_{\mathfrak{e}_i}(\boldsymbol{C})$ (Hill, 2008). This method thus transfers the information

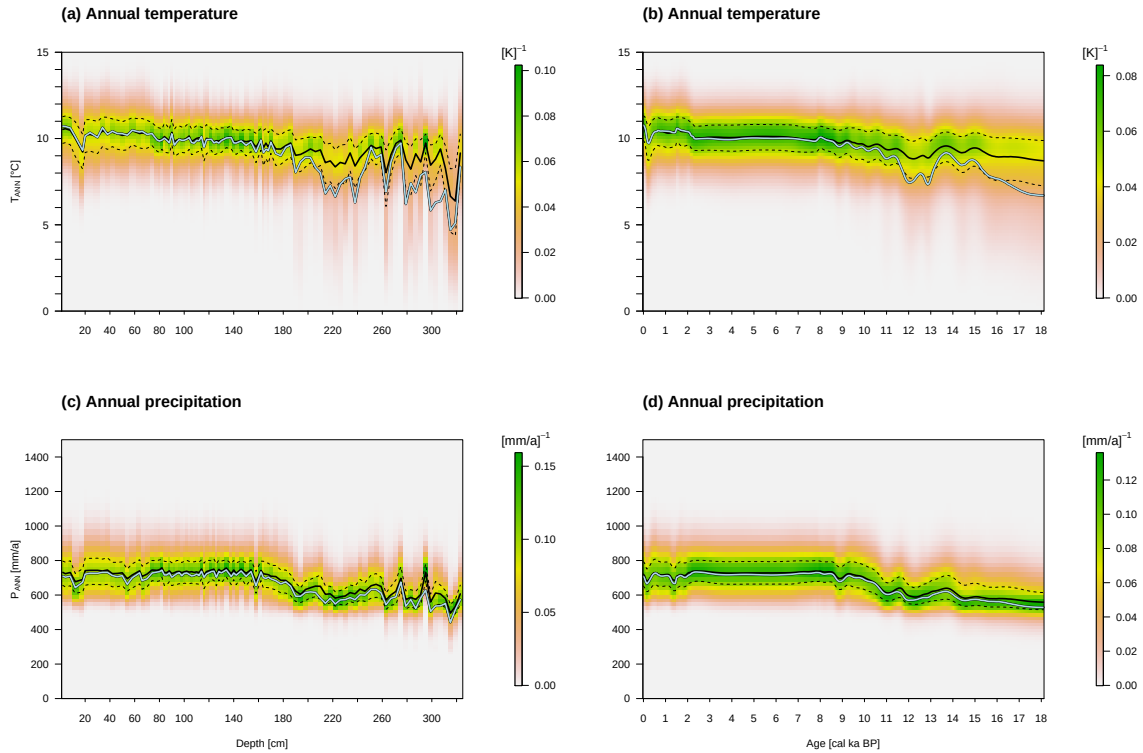


Figure 3.13: As Fig. 3.12, but the basis is the normalized product of independent posterior samples.

obtained during the MCMC simulation into the final result. Moreover, we can see the great similarity of both methods in terms of such a final result. A disadvantage of this combination is that the variance of $\mathbb{P}_{\text{mixt}}(\boldsymbol{C})$ is at least as large as the minimum of the variances of the input distributions $\mathbb{P}_{\mathfrak{C}_i}(\boldsymbol{C})$ (Hill, 2008). This leads to the high variances in Fig. 3.12, which make reasonable interpretation of paleoclimate difficult.

The next method remedies this effect, and the final result shown in Fig. 3.13 has a much lower uncertainty. This is achieved by calculating the normalized product of the individual input PDFs and is called conflation (Hill, 2008):

$$\mathbb{P}_{\text{confl}}(\boldsymbol{C}) = \frac{\prod_{i=1}^{N} \mathbb{P}_{\mathfrak{C}_i}(\boldsymbol{C})}{\int_{-\infty}^{\infty} \prod_{i=1}^{N} \mathbb{P}_{\mathfrak{C}_i}(\boldsymbol{C})\, d\boldsymbol{C}}.$$

At first glance, we see in panel (b) a big distinction between the black and the blue lines in the Last Glacial period. This is due to the fact that the conflation takes into account the respective accuracies of each $\mathbb{P}_{\mathfrak{C}_i}(\boldsymbol{C})$ by ultimately considering their overlapping regions (Hill, 2008). Therefore, this method is relatively sensitive to only comparatively small changes in the respective input PDFs, which has a corresponding effect on the final mean value.

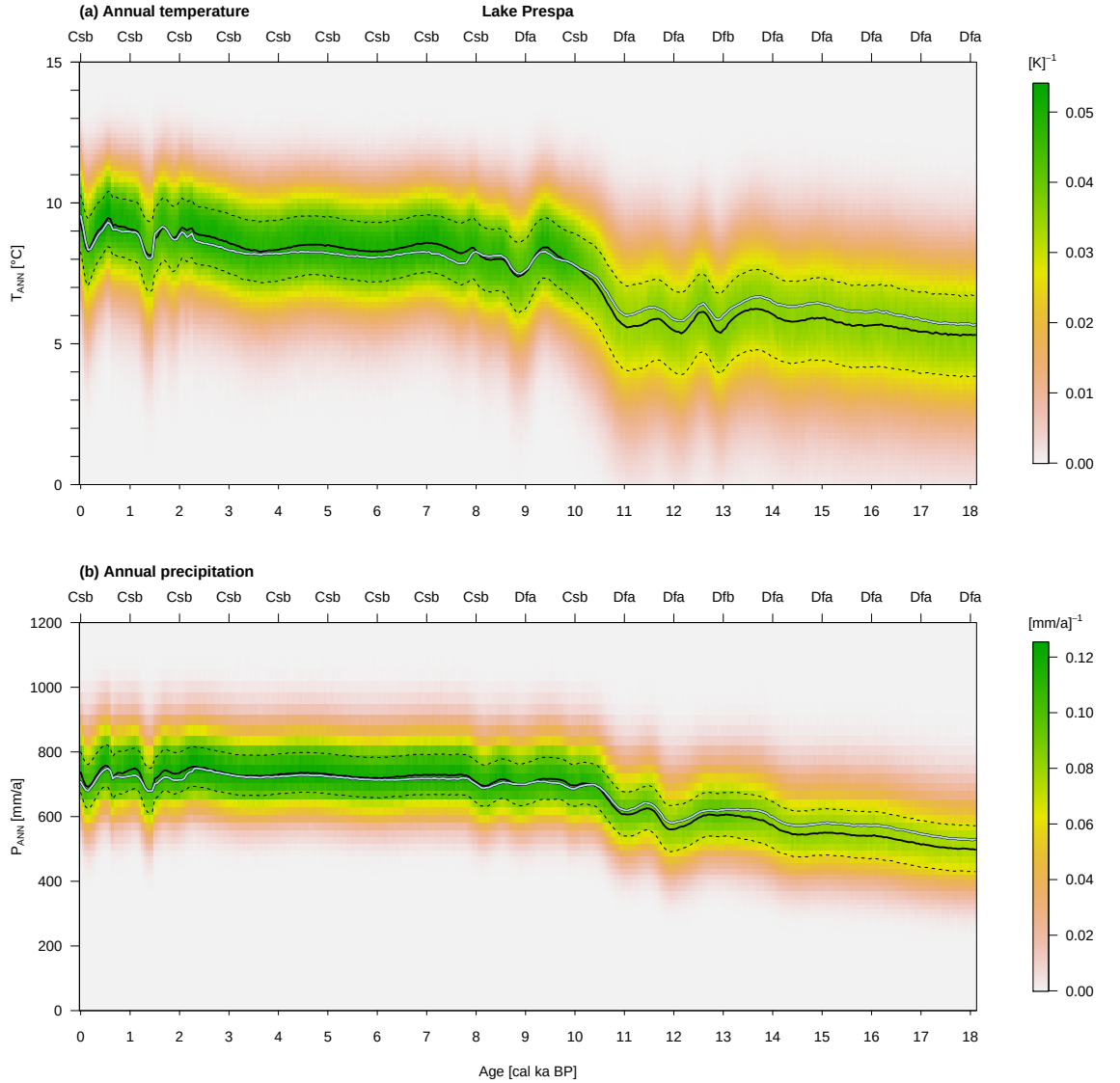


Figure 3.14: Posterior PDFs of annual temperature in (a) and annual precipitation in (b), based on the average of independent posterior random variables. In addition, the axis above the graph shows the approximated Koeppen-Geiger classifications corresponding to the PDFs for the respective periods.

The next method excludes the disadvantages mentioned above. That is, it reduces the variances due to the additional information of independent reconstructions and has the same average mean as the input distributions (Hill, 2008). A respective result is shown in Fig. 3.14 and is calculated via the mean of the input random variables: $Z_N = \frac{\mathfrak{C}_1+\mathfrak{C}_2+\ldots+\mathfrak{C}_N}{N} = \tilde{\mathfrak{C}}_1 + \tilde{\mathfrak{C}}_2 + \ldots + \tilde{\mathfrak{C}}_N$. Here the constant $1/N^{\frac{1}{\alpha}}$ is chosen with $\alpha = 1$. The latter is called the characteristic exponent of the stable density (Honerkamp, 2002). As a result, $1/N$ compresses the limit distribution around a point. This means that for larger

values of N a realization of Z_N scatters less and less around the mean of Z_N (Honerkamp, 2002), which explains the shape of the PDFs in Fig. 3.14. To calculate them, we use the fact that the PDF of the sum of the random variables can be obtained by calculating the convolution of the respective PDFs (cf. Eq. 2.2): $\mathbb{P}_{Z_N}(\boldsymbol{C}) = \mathbb{P}_{\tilde{\mathbf{c}}_1}(\boldsymbol{C}) * \mathbb{P}_{\tilde{\mathbf{c}}_2}(\boldsymbol{C}) * ... * \mathbb{P}_{\tilde{\mathbf{c}}_N}(\boldsymbol{C})$. Although more time-consuming to compute, all subsequent local climate reconstructions are based on this combination method because it has exactly the advantageous properties we want for such reconstructions.

At first glance, it can be seen that the course of the annual temperature in Fig. 3.14 (a) is similar to the reference curves in Fig. 3.8. This also affects the reconstructed annual precipitation, which is linked to T_{ANN} via the transfer functions and shows a similar pattern. It is also noticeable that both reconstructions (blue and black solid lines) give a very similar final result. We obtain the Koeppen-Geiger information by comparing the PDFs of the two climate variables with those of all Koeppen-Geiger classes. For this purpose, the so-called Earth Mover's Distance (EMD) is used, which takes into account the entire uncertainty structure (Rubner et al., 2000). The Koeppen-Geiger class with the lowest EMD is then used as the most likely. It should be noted that this estimate does not include direct intra-annual climate information. Nevertheless, we will see that this method gives usable results. In the following, the reconstruction in Fig. 3.14 is explained in more detail and compared with the corresponding palynological investigations by Panagiotopoulos et al., 2013 and Aufgebauer et al., 2012. For this purpose, we also predominantly use their temporal subdivision of vegetation history.

During the Late Pleniglacial, which in our case roughly corresponds to the period from **18 – 15 cal ka BP**, Panagiotopoulos et al., 2013 describe a relatively high AP (cf. Fig. 3.8 (b)) with low pollen influx. They indicate sparse vegetation, which Aufgebauer et al., 2012 describe with cold-tolerant herbs and pinus typical of stadial conditions. On average, our reconstructed variables show a temperature of about 6 °C and a precipitation of 600 mm/a, leading to the approximated Koeppen-Geiger climate Dfa, which describes a continental climate with hot summers without dry periods. The reconstructed values are similar to those in present-day areas in Russia between the Black Sea and Moscow, where the natural vegetation is described in Bohn et al., 2007 as a transition from forest steppes to herb-rich and pure feather grass steppes.

The period from **15 – 13.2 cal ka BP** encompasses the climatic fluctuations known as Bølling-Allerød, for which Panagiotopoulos et al., 2013 and Aufgebauer et al., 2012 describe a spread of tree taxa and a decline of herbaceous steppe taxa. From this they infer an increase in temperature and moisture availability. These fluctuations are also present in the two reconstructed variables. In the process, the mean temperature rises to 7 °C and precipitation by 50 mm/a. However, the Koeppen-Geiger class remains the same, indicating a relatively broad climate spectrum in Dfa.

During the Younger Dryas from ca. **13.2 – 11 cal ka BP**, vegetation reverts to near-glacial conditions, indicating a drier and colder climate (Panagiotopoulos et al., 2013; Aufgebauer et al., 2012). This is also reflected in the climate reconstruction, where temperatures are partly colder than in the lowest part of this core. Accordingly, the approximated Koeppen-Geiger classification changes to Dfb, which

describes a continental climate with warm summers without dry periods.

The early Holocene from ca. **11 – 7.9 cal ka BP** is described by Panagiotopoulos et al., 2013 and Aufgebauer et al., 2012 with an increase in vegetation density and the formation of closed oak forests. From this they deduce a steady increase in temperature and precipitation, which is reflected in our reconstruction. Here, T_{ANN} rises to around 9 °C and P_{ANN} to ca. 750 mm/a, leading to the climate classification Csb, which characterizes a Mediterranean climate with warm summers. In the period from about 9 – 7 cal ka BP, both variables show larger fluctuations that can be attributed to the 8.2 ka event. This Bond event can be seen well in Fig. 3.8 (b) and is described with a decrease in temperature and precipitation (Panagiotopoulos et al., 2013; Aufgebauer et al., 2012). However, our reconstruction shows an opposite behaviour in terms of temperature. This is because, for example, Ulmus is weighted higher due to the approximations of the reference curves and the climate anchor points. The corresponding transfer function shows a relatively cold and humid climate, which together with a pollen fluctuation in the period around the 8.2 ka event leads to this reconstructed pattern.

In the middle and late Holocene from about **7.9 cal ka BP onwards**, the vegetation shows a rapid spread of several thermophilous trees (Panagiotopoulos et al., 2013) and thus a return to the Mediterranean climate with warm summers (Csb). From 1.9 cal ka BP we see in Fig. 3.8 (b) strong fluctuations in the pollen spectrum and a decrease in AP. This is due to the increasing influence of humans on vegetation (Panagiotopoulos et al., 2013), which is also partly reflected in our reconstruction. Both climate variables, however, do not show a steady decline as the reconstructions of the most recent period are drawn towards the climate anchor points (cf. Fig. 3.7). Furthermore, the a priori fixed parameters of the beta distributions prevent too close approximation to the reference curves. The recent Koeppen-Geiger classification of Lake Prespa does not correspond to what is seen in our reconstruction. Nevertheless, we will see in Sect. 3.3.6 that the neighbouring Lake Ohrid, which has a similar climate, is referred to as Csb. This means that this area is a transition zone between Cfb and Csb, which confirms the method for determining the approximated Koeppen-Geiger climate classification as useful.

With the exception of the period from 9 – 7.9 cal ka BP, our reconstruction agrees with Panagiotopoulos et al., 2013 and Aufgebauer et al., 2012. This becomes particularly clear when looking at the Koeppen-Geiger climate classification of the respective periods. This allows them to be compared well with present-day areas and their vegetation, giving a more comprehensive picture of climate reconstruction.

Since it is quite time-consuming to obtain the above AP/NAP curves for most sites from the EPD, we introduce another reference curve based on the results of Giesecke et al., 2017. They provide reconstructed European maps for 16 taxa based on the EPD datasets. These show the relative occurrence of these plants in relation to different thresholds in the period 0 – 15 cal ka BP. The temporal resolution is 500 years, during which we obtain information on regional occurrence (threshold 1) to regional dominance (threshold 4) of each taxa. Giesecke et al., 2017 finally interpolate these

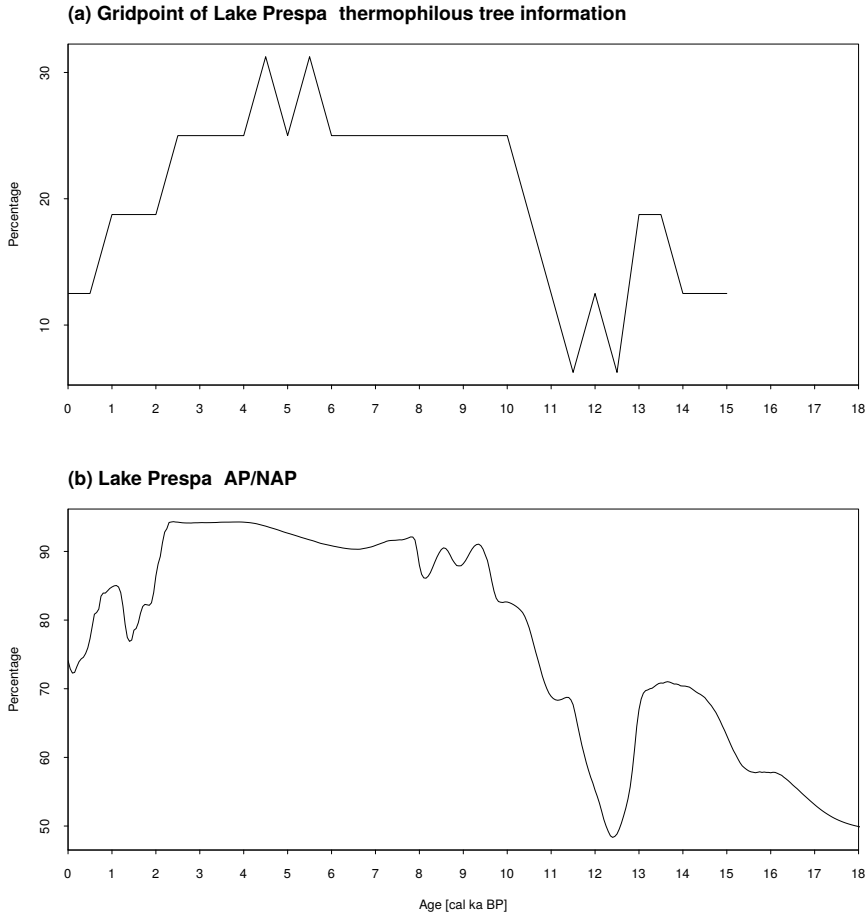


Figure 3.15: Reference curves based on (a) thermophilous tree information from Giesecke et al., 2017 and (b) on total terrestrial pollen sums: arboreal (AP) and non-arboreal (NAP) pollen taxa from Lake Prespa (Panagiotopoulos et al., 2013).

results to a $0.5^\circ \times 0.5^\circ$ grid. In this way, we can retrieve this information for the respective lakes and mires. For Lake Prespa, the grid point closest to the lake is considered and all values above the regional occurrence are taken into account. If only values above threshold 4 were considered, part of the reference curve would be equal to 0, as the regional dominance of these taxa is missing (not shown). We consider 7 different thermophilous trees, which are compared with the other 9 taxa, giving us a relative abundance of these plants that varies according to regional occurrence and regional dominance. The result for Lake Prespa is shown in Fig. 3.15 (a). When comparing with the AP/NAP curve in (b), one immediately notices a great similarity. The main difference is in the respective percentages, temporal resolution, and restriction to the period from $0 - 15$ cal ka BP. The latter has the greatest impact on climate reconstruction, as shown in Fig. 3.16. Here the blue line shows the final reconstruction based on the Giesecke reference curve. Despite the high degree of resemblance between this and the black solid line with respect to both climate variables, P_{ANN} shows

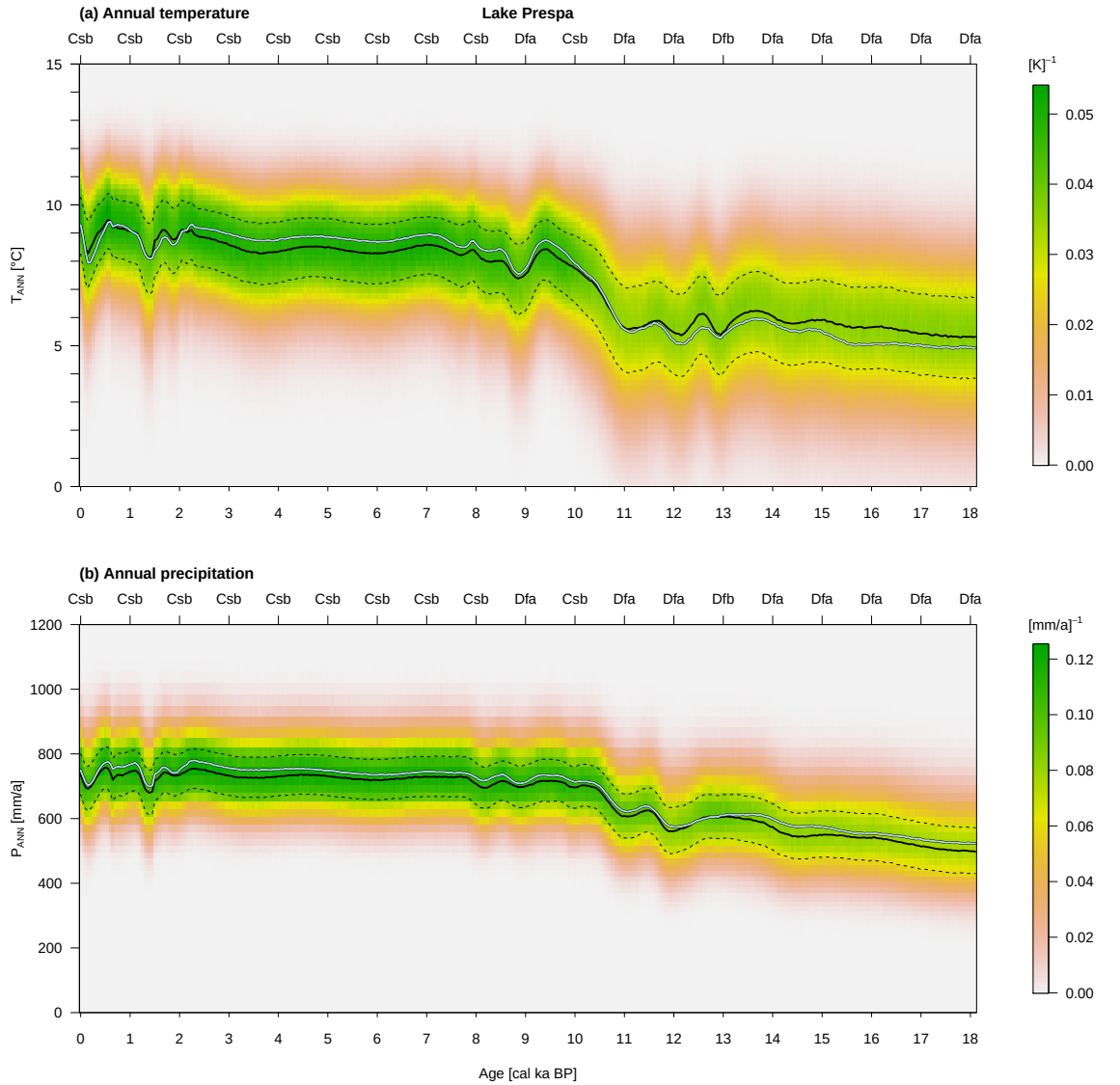


Figure 3.16: As Fig. 3.14, but the blue line shows the final reconstruction based on the Giesecke reference curve.

greater differences in the lower core, which is also observed in T_{ANN}. However, large fluctuations are also observed in (a) during the middle Holocene. These are directly related to those between 15-18 cal ka BP, as the corresponding reconstruction samples for the entire period are generated at once. So, for example, if the blue line in the Late Pleniglacial has a colder temperature, this may affect the entire subsequent course. All in all, Fig. 3.16 shows that a reasonable climate reconstruction can be determined under the aspects just described.